coursera

Gemini Capstone Project ✕

Today's goals

Complete any 3 learning items · 3/3

○ Complete 3 readings · 1/3

Watch a video

Module 1

Function Calling and Agents

✓ Course introduction
Reading • 10 min

✓ From static to dynamic AI
Video • 2 min

✓ Function calling fundamentals
Video • 6 min

✓ Building AI agents
Video • 4 min

✓ Activity: AI agent architecture design
Practice Assignment • Grade: 100%

✓ Evaluating AI agent architecture designs
Dialogue • 15 min

✓ Knowledge check: Building your AI agent
Practice Assignment • Grade: 100%

✓ Conclusion and next steps
Reading • 10 min

Module 2

Production Planning and Deployment

✓ Planning to launch your AI application
Video • 2 min

✓ Understanding tokens and costs
Video • 4 min

○ **Easy deployment and monitoring**
Reading • 10 min

○ From code to cloud with ease
Video • 4 min

○ Activity: Deploy your AI-powered application
Practice Assignment • 1h

○ The AI application production process
Dialogue • 15 min

○ Production planning and deployment
Graded Assignment • 30 min

○ Conclusion and next steps
Reading • 10 min

Easy deployment and monitoring

0:06 / 10:38 1x

Bringing your AI application to life

Moving an application from a prototype in Google AI Studio to a live product is a major milestone. You've refined your prompts, tested ideas, and selected the right models. Now, you're ready to share your creation with the world.

This transition—from a controlled test environment to real-world use—brings new considerations. Success depends not only on innovation but also on building an application that is reliable, efficient, and sustainable. Even a powerful tool will struggle if it's costly to run or prone to errors.

This reading will guide you through the process of moving your AI Studio project into production. You'll learn how to plan for real-world use by managing costs, deployment with Google Cloud Run, and performance monitoring to maintain stability and value for users.

Together, these three connected steps—planning, deployment, and monitoring—form the foundation for creating AI applications that are innovative, practical, and built for long-term success.

Understanding and managing costs

Before deploying your application, the first and most essential step is planning—that begins with understanding the costs involved. In AI development, expenses can accumulate quickly if you do not analyze how the model processes data. The main cost when using the Gemini API comes from token usage.

A token is a small unit of text the model reads or generates. Each prompt and response is broken into tokens—essentially the digital currency that powers the model's work. Each time your application sends or receives text, it spends tokens. To create a sustainable and affordable product, you must become skilled at tracking and managing this usage.

Estimating usage and projecting costs

Your first task is to understand how many tokens your application typically consumes. AI Studio makes this process simple by displaying the token count for every request you send. Begin by testing a few sample interactions that represent typical user behavior. For instance, you might try short factual prompts, longer creative ones, and a few edge cases to capture variation in response length.

To project overall costs, estimate three key factors:

1. **Average tokens per request:** Run several trials in AI Studio and note the token usage for both input and output. Averaging across test cases will give you a realistic estimate.
2. **Estimated number of users:** Consider how many people will use your application daily and how many times each might interact with it. Even small assumptions here can greatly affect cost projections.
3. **Cost per token:** Each Gemini model has its own pricing. Gemini Flash offers faster and lower-cost performance, while Gemini Pro delivers deeper reasoning and more advanced outputs at a higher price. The exact pricing can be reviewed on the Google Cloud website.

When you multiply the average tokens per request by the estimated number of daily requests and then apply the cost per token, you'll have a projected daily or monthly expense. This turns vague uncertainty into measurable numbers you can plan for. Predictable costs help you manage budgets, especially as your user base grows.

Controlling and optimizing costs

Once you understand the baseline, the next step is to control your costs strategically. The most effective lever is model selection. If your application focuses on straightforward tasks—like generating summaries, short answers, or lists—then a lightweight model such as Gemini Flash might be ideal. For applications that require advanced reasoning or complex decision-making, a more capable model may justify the higher cost.

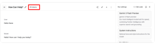
Balancing quality, speed, and cost is an ongoing process. A higher-tier model is not always the best option if its price makes your application difficult to sustain. Experiment with multiple models and measure both output quality and efficiency before finalizing your choice.

Additional cost-saving approaches include:

- Refining your prompts to be concise and targeted rather than open-ended.
- Limiting the maximum output length to prevent unnecessary text generation.
- Caching or reusing responses where possible to reduce repeated requests.
- Monitoring usage trends to identify where optimization will have the most impact.

By mastering this stage of planning—understanding token usage, analyzing trade-offs, and adopting optimization practices—you lay the groundwork for sustainable AI applications that can operate efficiently while continuing to deliver high-value results for your users.

Token count in Google AI Studio



User
Hello there

Model
Hello! How can I help you today?

Gemini 3 Flash Preview
gemini-3-flash-preview
Our most intelligent model built for speed, combining frontier intelligence with superior search and grounding.

System instructions
Optional tone and style instructions for the model

Deploying your application with Google Cloud Run

Once you have a clear plan for managing costs, the next step is deployment—making your AI application available for users to access online. Deployment marks the point where your work moves from a private test environment to a live, interactive product.

In the past, this process required setting up and maintaining servers, ensuring uptime, and handling complex configurations. Today, tools like Google Cloud Run make deployment much simpler by managing the infrastructure for you.

Understanding serverless deployment

Cloud Run is a *serverless* platform, which means you don't need to set up or manage any physical or virtual servers. You focus on your application code, and Cloud Run automatically runs it in a secure, scalable environment. When traffic increases, Cloud Run scales up to handle the load; when activity slows, it scales down—sometimes to zero—so you only pay for the resources used.

This automatic scaling makes Cloud Run cost-efficient and reliable, especially for applications with varying or unpredictable user traffic. Whether your app serves a handful of users or thousands, Cloud Run adjusts seamlessly without manual intervention.

Moving from prototype to production

AI Studio integrates directly with Cloud Run, making it easy to move from a working prototype to a production-ready application. Once your prototype is finalized, AI Studio can generate the code and prepare it for deployment.

This code is packaged into a container, which includes everything your application needs to run—its code, libraries, and dependencies. Containers make deployment consistent: your app behaves the same way on your local machine, in testing, or in the cloud.

You can think of a container as a ready-to-ship package for your software. Once it's built, Cloud Run handles the setup automatically. You upload your container, and the service launches it, assigning a unique URL so your application becomes instantly accessible to users.

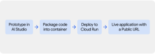
Securing your deployment

A key part of deployment is security, particularly in managing your API key. The API key allows your code to connect securely to the Gemini API. Never embed this key directly in your code—doing so can expose it to others, leading to unauthorized access and additional costs.

Instead, use environment variables to store sensitive credentials securely outside your code. When deploying to Cloud Run, you can configure an environment variable for your API key so your application reads it safely at runtime. This method keeps your key protected and allows you to update it easily without changing or redeploying your code.

By using Cloud Run's streamlined deployment process and secure configuration options, you can confidently take your AI Studio project live—focusing on performance and user experience rather than technical setup or maintenance.

From prototyping to a publicly accessible application



Prototype in AI Studio
Package code into container
Deploy to Cloud Run
Live application with a Public URL

Monitoring your live application for continued success

Launching your application is just the beginning. To keep it reliable, efficient, and valuable to users, you must monitor it regularly. Monitoring means tracking how your application performs in real-world conditions and using that data to make informed improvements. Google Cloud provides tools that integrate with Cloud Run to help you track usage, performance, and costs effectively.

Tracking usage and performance

The first focus area is usage and performance monitoring. Understanding when users are most active and how your application responds helps you plan updates and manage performance. If usage spikes at certain times, you can prepare for higher demand or schedule updates during quieter periods.

Cloud Run automatically generates detailed logs that record your app's activity. Reviewing these logs helps you identify slow responses or errors early, so you can fix issues quickly and keep performance steady.

Monitoring and controlling costs

The second focus area is cost monitoring. Even with careful planning, expenses can fluctuate once your app goes live. The Google Cloud Console shows real-time usage and spending data, and you can set billing alerts to notify you when costs approach a set limit.

These alerts act as an early warning system, preventing unexpected bills and helping you stay within budget as your application scales.

Maintaining and optimizing your application

The third area is maintenance and optimization. Monitoring data reveals where improvements are needed—new features, streamlined workflows, or updated models. Maintenance also means keeping dependencies current and exploring newer, more cost-efficient models when available.

By combining these practices, you move from reacting to issues after they occur to anticipating them before they affect users. This proactive approach helps your application stay stable, affordable, and ready to grow.

Key takeaways

Ongoing monitoring keeps your application stable, cost-efficient, and aligned with user needs. By using Google Cloud's integrated tools, you can track performance, detect issues early, and control spending through real-time insights and billing alerts. Regular maintenance and optimization help your application stay up to date, scalable, and ready to evolve as technology and user expectations change.

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